

FIFO Depth Analysis

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1. Given Data and Method

The analysis considers a FIFO connecting a writing module (Module A) to a reading module (Module B). The parameters given in the problem are

$$f_A = 100 \text{ MHz}, \quad f_B = 40 \text{ MHz}, \quad B = 100 \text{ data items.}$$

Three operating cases are considered:

1. No idle cycles between writes or reads.
2. Two idle cycles between successive writes and two idle cycles between successive reads.
3. Two idle cycles between successive writes and one idle cycle between successive reads.

The required FIFO depth is the maximum number of unread data items that accumulate in the FIFO during the transfer.

For a clock frequency f ,

$$T = \frac{1}{f}.$$

If there are I idle cycles between two successive operations, one operation takes $(I + 1)$ clock cycles. Therefore,

$$T_{\text{eff}} = (I + 1)T$$

and

$$f_{\text{eff}} = \frac{f}{I + 1}.$$

When the effective write rate is higher than the effective read rate, the FIFO must absorb the difference.

2. Example 1: No Idle Cycles

For the first case,

$$I_W = 0, \quad I_R = 0.$$

The write and read clock periods are

$$T_A = \frac{1}{100 \text{ MHz}} = 10 \text{ ns}$$

and

$$T_B = \frac{1}{40 \text{ MHz}} = 25 \text{ ns.}$$

Since there are no idle cycles,

$$T_{W,\text{eff}} = 10 \text{ ns}, \quad T_{R,\text{eff}} = 25 \text{ ns.}$$

The 100-item write burst takes

$$T_{\text{burst}} = 100(10) = 1000 \text{ ns.}$$

During this interval, the reader can process

$$N_R = \frac{1000}{25} = 40$$

data items.

Therefore, the maximum accumulated data is

$$D = 100 - 40 = \boxed{60 \text{ entries}}.$$

Hence,

$$\boxed{D_1 = 60}$$

is the required FIFO depth for Example 1.

3. Example 2: Two Idle Cycles on Both Sides

In this case,

$$I_W = 2, \quad I_R = 2.$$

There are three clock cycles per actual write or read operation:

$$I + 1 = 3.$$

Thus, the effective write period is

$$T_{W,\text{eff}} = 3(10) = 30 \text{ ns},$$

and the effective read period is

$$T_{R,\text{eff}} = 3(25) = 75 \text{ ns}.$$

Equivalently, the effective frequencies are

$$f_{W,\text{eff}} = \frac{100}{3} = 33.33 \text{ MHz}$$

and

$$f_{R,\text{eff}} = \frac{40}{3} = 13.33 \text{ MHz}.$$

The time required for the 100 writes is

$$T_{\text{burst}} = 100(30) = 3000 \text{ ns}.$$

The number of reads possible during this interval is

$$N_R = \frac{3000}{75} = 40.$$

Therefore,

$$D = 100 - 40 = \boxed{60 \text{ entries}}.$$

Hence,

$$\boxed{D_2 = 60}.$$

Although idle cycles were introduced on both sides, they are equal. Consequently, both effective rates are reduced by the same factor of three, so their ratio remains unchanged and the required FIFO depth remains 60 entries.

4. Example 3: Two Write Idle Cycles and One Read Idle Cycle

For the third case,

$$I_W = 2, \quad I_R = 1.$$

The write side therefore requires

$$I_W + 1 = 3$$

cycles per write, while the read side requires

$$I_R + 1 = 2$$

cycles per read.

The effective write period is

$$T_{W,\text{eff}} = 3(10) = 30 \text{ ns},$$

so

$$f_{W,\text{eff}} = \frac{100}{3} = 33.33 \text{ MHz}.$$

The effective read period is

$$T_{R,\text{eff}} = 2(25) = 50 \text{ ns},$$

so

$$f_{R,\text{eff}} = \frac{40}{2} = 20 \text{ MHz}.$$

The 100 writes require

$$T_{\text{burst}} = 100(30) = 3000 \text{ ns}.$$

During this interval, the reader can process

$$N_R = \frac{3000}{50} = 60$$

data items.

Therefore,

$$D = 100 - 60 = \boxed{40 \text{ entries}}.$$

Hence,

$$\boxed{D_3 = 40}.$$

5. General Formula

For a burst containing B data items, with I_W idle cycles between writes and I_R idle cycles between reads,

$$f_{W,\text{eff}} = \frac{f_W}{I_W + 1}$$

and

$$f_{R,\text{eff}} = \frac{f_R}{I_R + 1}.$$

The number of data items read while the complete write burst is being transferred is

$$N_R = B \frac{f_{R,\text{eff}}}{f_{W,\text{eff}}}.$$

Therefore, when the writer is faster,

$$D = B - N_R$$

which gives

$$\boxed{D = B \left(1 - \frac{f_R(I_W + 1)}{f_W(I_R + 1)} \right)}.$$

For the three cases:

$$D_1 = 100 \left(1 - \frac{40}{100} \right) = \boxed{60},$$

$$D_2 = 100 \left[1 - \frac{40(3)}{100(3)} \right] = \boxed{60},$$

and

$$D_3 = 100 \left[1 - \frac{40(3)}{100(2)} \right] = \boxed{40}.$$

6. Final Results

Example	I_W	I_R	Required FIFO Depth
1	0	0	$\boxed{60}$ entries
2	2	2	$\boxed{60}$ entries
3	2	1	$\boxed{40}$ entries

7. Conclusion

The FIFO depth is determined by the maximum amount of data written before the corresponding data can be consumed by the read side.

For the first two cases, the relative write/read rate is unchanged, so the required depth is

$$\boxed{60 \text{ entries}}.$$

In the third case, the read side has fewer idle cycles than the write side. Consequently, the reader operates relatively faster and the amount of accumulated data decreases to

$$\boxed{40 \text{ entries}}.$$

Therefore, the final answers are

$$\boxed{D_1 = 60, \quad D_2 = 60, \quad D_3 = 40}.$$